

Patuakhali Science and Technology University  
 B. Sc. in CSE (Engg.) L-2, S-II Class Test 2022 (July-December)  
 Course Code: AES-221 Course Title: Government and Economics  
 Credit Hour: 3.00 Full Marks: 15 Time: 55 minutes  
 Answer all the questions.

Marks

1. a. Define politics & government. price 2  
 b. Mention three definitions about modern political science. 2  
 c. Briefly discuss the nature of political science. 3.5  
 2. a. Define economics and discuss the subject matter of economics. 3.5  
 b. What is demand curve? Derive the individual demand curve with the help of a hypothetical demand schedule. 4
- 5, 7, 10, 15, 20, 30
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Patuakhali Science and Technology University

B.Sc. in CSE 4th Semester (L-2, S-11) Class Test (July-December 2021)  
 Course Code: RSL 211 Course Title: Government and Economics  
 Credit Hour: 3.0 Full Marks: 20 Duration: 55 Minutes

1. a) Diagrammatically show the consumer's behavior analysis. State the relationship between total utility and marginal utility showing their formula. 5  
 b) Distinguish between: 5
- i) Production and consumption,
  - ii) Assumption and hypothesis
  - iii) Free goods and economic goods
  - iv) Value and price
2. Mention the traditional and modern definition of political science? 5  
 3. Illustrate the scope of political science. 5
4. a. Define economics. Distinguish between micro economics and macro economics. 3  
 b. What is utility? Discuss the different types of utility with example.  
 c. Differentiate between:

Patuakhali Science and Technology University

Dept. of Economics and Sociology

B. Sc. Engg. (CSE) 4<sup>th</sup> Semester (L-2, S- II) Final Examination 2022 (July-December)

Course Code: AES 221 Course Title: Government and Economics

Credit Hour: 3.00

Full Marks: 70

Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 5(five) of the following questions.

1. a Describe the scope of political science. 4  
b What is political ideology? Make a list about the various political ideologies of modern political systems. 5  
c What do you mean by political authority? Prepare a comparison table according to different types of authority. 5
2. a Briefly discuss about the seven characteristics of nation. 4  
b Narrate the sources of political legitimacy. 6  
c What are the eight elements or the roots of nationalism? 4
3. a Differentiate the main dimensions of government and governance. 6  
b What are the differences between power and authority? 4  
c Discuss the elements of state. sovereignty, territory, population, government. 4
4. a Define economics. Distinguish between micro economics and macro economics. 3  
b What is utility? Discuss the different types of utility with example. 3  
c Differentiate between: 8  
i) Free goods and Economic goods  
ii) Private goods and public goods  
iii) Theory and Law  
iv) Value and Price
5. a What is law of demand? Explain the law of demand with assumptions and limitations. 4  
b Discuss the non-price factors that affect demand. 4  
c Explain the law of equi-marginal utility with assumptions and limitations. 6
6. a Define supply. Graphically explain the contraction and extension in supply and decrease and increase in supply. 6  
b What is elasticity of supply? Graphically explain the types of elasticity of supply. 4  
c Suppose the income of a person is Tk. 40,000 per month and he purchases ten CD's per year. Let us assume that the monthly income of the consumer increases 15% and the quantity demanded of CD's per year rises to 5%. Calculate the income elasticity of demand for CD's. 4

original quantity(Q): 10 CD's  
Income(Y): 40,000

$$\Delta Q = 10 + 10 \times \frac{5}{100} - 10 = 0.5$$

(Figures in the right margin indicate full marks. Split answering of any question is not recommended.)  
 Answer any 5(five) of the following questions.

- 1.a Define government and politics. 4  
 b Narrate the scope of political science. 4  
 c Differentiate between the government and governance. 6
- 2.a Discuss about the elements of state. 7  
 b Narrate the theories of origin of state. 7
- 3.a Illustrate the traditional and modern form of governments. Which form of government is the best from according to Aristotle and why? 7  
 b Make a comparison table about the various types of authorities. 7
- 4.a What is economics? Differentiate between micro economics and macro economics. 3  
 b What is utility? Discuss the different types of utility with example. 3  
 c Differentiate between:  
 i) Assumption and hypothesis  
 ii) Private goods and public goods  
 iii) Theory and Law  
 iv) Value and Price
- 5.a What is law of demand? Explain the law of demand with assumptions and limitations. 4  
 b Discuss the non-price factors that affect supply. 4  
 c Explain the law of diminishing marginal utility with assumptions and limitations. 6
- 6.a Define supply. Graphically explain the contraction and extension in supply, and decrease and increase in supply. 6  
 b What is elasticity of supply? Graphically explain the types of elasticity of supply. 4  
 c Suppose the income of a person is Tk. 30,000 per month and he purchases ten CD's per year. Let us assume that the monthly income of the consumer increases 10% and the quantity demanded of CD's per year rises to 5%. Calculate the income elasticity of demand for CD's. 4

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Income  
 Price  
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1. a) What is political ideology? 3.0  
 b) What do you mean by authority? State the comparative picture between traditional, charismatic and legal rational authority. 6.0  
 c) Briefly discuss about the seven characteristics of a nation. 5.0
  2. a) What are the differences between power and authority? 4.0  
 b) Differentiate the main dimensions of government and governance. 5.0  
 c) Narrate the sources of political legitimacy. 5.0
  3. a) What are the eight elements or the roots of nationalism? 4.0  
 b) Mention the types of traditional and modern government according to Aristotle and Leacock. 5.0  
 c) Discuss the functions of legislature. 5.0
  4. a) Define Indifference curve with its assumptions. 4.0  
 b) State the characteristics of an indifference curve with necessary figures. 4.0  
 c) Define budget line. Suppose that the consumer has a Budget of Tk. 100. The Price of Mango is Tk. 20 and Price of Apple is Tk.10. Draw a budget line with these parameters. 3.0  
 d) Explain the consumer's equilibrium with indifference curve & budget line. 3.0
  5. a) Define supply and law of supply. 3.0  
 b) Differentiate between supply schedule and supply curve with figure. 3.0  
 c) Suppose that an individual's demand function for commodity X is  $Q_{d_x} = 8 - 2P_x$ . By substituting various prices of X into this demand function, we get the individual's demand schedule shown in the Table. Prepare an individual demand curve from the schedule. 2.0
- |             |   |   |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|---|---|
| $P_x$ (Tk.) | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| $Q_{d_x}$   | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
- d) Explain the factors affecting demand and supply of a commodity with examples. 6.0
  6. a) Differentiate between price elasticity and income elasticity of demand. 3.0  
 b) Discuss the numerical method of measuring price elasticity of demand with mathematical example in each case. 5.0  
 c) Classify goods on the basis of income elasticity of demand. 2.0  
 d) When income of a consumer is Tk. 4000 and his demand of a product is 100 kg. If income rises to Tk. 5000 its demand falls to 80 kg. Calculate the income elasticity and comment on the type of goods. 4.0

**Patuakhali Science and Technology University**

B.Sc. Engg. in CSE 4<sup>th</sup> Semester (L-2, S-II) Final Examination-2020 (July-December)

Course Code: AES 221

Course Title: Government and Economics

Credit Hour: 3.0

Full Marks: 70

Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering is not recommended.]

**Answer any five of the following questions.**

1. a) Discuss the five principles of governance. 4.0  
b) Explain the differences between governance and government. 4.0  
c) Interpret the basic principles of democracy. 6.0
2. a) Sketch the types of authority. 4.0  
b) Illustrate the factors that determine legitimacy. 4.0  
c) Narrate the functions of legislature. 6.0
3. a) State the four theories of origin of the state. 3.0  
b) Briefly discuss the elements of the state. 6.0  
c) Describe the necessity of state. 5.0
4. a) Define different types of elasticity of demand. Classify goods on the basis of different elasticity of demand. 5.0  
b) Write down different methods of measuring price elasticity of demand. Explain the numerical method with graphical views. 5.0  
c) Suppose that price of a commodity is tk. 4.00 and its demand is 100 kg. If price falls to tk 3.00 its demand rises to 130 kg. Calculate the price elasticity of demand. 4.0
5. a) What do you mean by production and production function? 3.0  
b) Discuss the characteristics of different stages of a typical production function. In which stage a rational manager should operate and why? 6.0  
c) What should be the exact profit point? Explain with mathematical data that profit maximization is attained at that level of input use where  $MVP = MFC$ . 5.0
6. a) What is meant by explicit cost, implicit cost and opportunity cost? 4.0  
b) Suppose you are running a small business and incur the following cost: labor = Tk. 70,000; raw materials = Tk40,000; Interest on bank loan = Tk 3,000. You are not paying explicit rent, because you own the building where you are operating the business. However, if you would rent it out, you could be earning Tk. 20,000. You also estimate the opportunity cost of your self-engagement in the firm to be worth Tk. 30,000. Moreover, you suppose that the company's total revenue is Tk280,000. Calculate the explicit cost, implicit cost and economic cost, accounting profits and economic profits? 6.0  
c) Explain seven cost concepts with graphical presentation and critical explanations. 4.0

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